

GUJARAT TECHNOLOGICAL UNIVERSITY
BE –SEMESTER 1&2(NEW SYLLABUS)EXAMINATION- WINTER 2018

Subject Code: 3110014**Date: 07-01-2019****Subject Name: Mathematics - I****Time: 10:30 am to 01:30 pm****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

	Marks
Q.1 (a) State Cayley– Hamilton theorem. Find eigen values of A and A^{-1} , where $A = \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}$	03
(b) State L' Hospital's Rule. Use it to evaluate $\lim_{x \rightarrow 0} \left[\frac{1}{x^2} - \frac{1}{\sin^2 x} \right]$	04
(c) Investigate convergence of the following integrals:	07
(i) $\int_5^{\infty} \frac{5x}{(1+x^2)^3} dx$	
(ii) $\int_0^{\infty} \frac{x^{10}(1+x^5)}{(1+x)^{27}} dx$	
Q.2 (a) Test the convergence of series $\sum_{n=1}^{\infty} \frac{4^n + 5^n}{6^n}$	03
(b) State the p-series test. Discuss the convergence of the series $\sum_{n=2}^{\infty} \frac{n+1}{n^3 - 3n + 2}$	04
(c) State D'Alembert's ratio test and Cauchy's root test. Discuss the convergence of the following series:	07
(i) $\sum_{n=1}^{\infty} \frac{4^n (n+1)!}{n^{n+1}}$	
(ii) $\sum_{n=2}^{\infty} \frac{n}{(\log n)^n}$	
OR	
(c) Test the convergence of the series $\frac{1}{1 \cdot 2 \cdot 3} + \frac{x}{4 \cdot 5 \cdot 6} + \frac{x^2}{7 \cdot 8 \cdot 9} + \frac{x^3}{10 \cdot 11 \cdot 12} + \dots$; $x \geq 0$	07
Q.3 (a) Reduce matrix $A = \begin{bmatrix} 1 & 5 & 3 & -2 \\ 2 & 0 & 4 & 1 \\ 4 & 8 & 9 & -1 \end{bmatrix}$ to row echelon form and find its rank.	03
(b) Derive half range sine series of $f(x) = \pi - x$, $0 \leq x \leq \pi$	04
(c) Find the eigen values and corresponding eigen vectors for the matrix A where $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$	07

OR

- Q.3** (a) Expand $e^{x \sin(x)}$ in power of x up to the terms containing x^6 . **03**
(b) Solve system of linear equation by Gauss Elimination method, if solution exists. **04**
 $x + y + 2z = 9$; $2x + 4y - 3z = 1$; $3x + 6y - 5z = 0$
(c) Find Fourier series of $f(x) = \begin{cases} x, & -1 < x < 0 \\ 2, & 0 < x < 1 \end{cases}$ **07**
- Q.4** (a) Discuss the continuity of the function f defined as **03**
 $f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2}; & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$
(b) Define gradient of a function. Use it to find directional derivative of $f(x, y, z) = x^3 - xy^2 - z$ at $P(1, 1, 0)$ in the direction of $\vec{a} = 2\hat{i} - 3\hat{j} + 6\hat{k}$. **04**
(c) Find the shortest and largest distance from the point $(1, 2, -1)$ to the sphere $x^2 + y^2 + z^2 = 24$. **07**

OR

- Q.4** (a) Find the extreme values of $x^3 + 3xy^2 - 3x^2 - 3y^2 + 4$ **03**
(b) Evaluate $\int_0^2 \int_0^{\sqrt{4-x^2}} (x^2 + y^2) dy dx$ by changing into polar coordinates. **04**
(c) (i) If $u = x^2y + y^2z + z^2x$ then find out $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$ **07**
(ii) If $x^3 + y^3 = 6xy$ then find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$
- Q.5** (a) Evaluate $\iint_R y \sin(xy) dA$, where R is the region bounded by $x = 1$, $x = 2$, $y = 0$ and $y = \frac{\pi}{2}$. **03**
(b) By changing the order of integration, evaluate $\int_0^3 \int_y^3 \frac{xdxdy}{x^2 + y^2}$ **04**
(c) Find the volume below the surface $z = x^2 + y^2$, above the plane $z = 0$, and inside the cylinder $x^2 + y^2 = 2y$. **07**

OR

- Q.5** (a) Evaluate integral $\iint_R \frac{rdrd\theta}{\sqrt{a^2 + r^2}}$ over the region R which is one loop of $r^2 = a^2 \cos 2\theta$ **03**
(b) Evaluate the integral $\int_1^e \int_1^{\log y} \int_1^{e^x} (x^2 + y^2) dz dx dy$. **04**
(c) Find the volume of the solid obtained by rotating the region R enclosed by the curves $y = x$ and $y = x^2$ about the line $y = 2$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER-2019****Subject Code: 3110014****Date: 06/06/2019****Subject Name: Mathematics – I****Time: 10:30 AM TO 01:30 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

- | | Marks |
|--|-----------|
| Q.1 (a) Use L'Hospital's rule to find the limit of $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\ln x} \right)$. | 03 |
| (b) Define Gamma function and evaluate $\int_0^\infty e^{x^2} dx$. | 04 |
| (c) Evaluate $\int_0^3 \int_{\frac{1}{\sqrt{x}}}^1 e^{y^2} dy dx$. | 07 |
| Q.2 (a) Define the convergence of a sequence (a_n) and verify whether the sequence whose n^{th} term is $a_n = \left(\frac{n+1}{n-1} \right)^n$ converges or not. | 03 |
| (b) Sketch the region of integration and evaluate the integral $\iint_R (y - 2x^2) dA$ where R is the region inside the square $ x + y = 1$. | 04 |
| (c) (i) Find the sum of the series $\sum_{n \geq 2} \frac{1}{4^n}$ and $\sum_{n \geq 1} \frac{4}{(4n-3)(4n+1)}$. | 07 |
| (ii) Use Taylor's series to estimate $\sin 38^\circ$. | |
| OR | |
| (c) Evaluate the integrals $\int_0^\infty \int_0^\infty \frac{1}{(1+x^2+y^2)^2} dx dy$ and $\int_0^1 \int_{\sqrt{z}}^1 \int_0^{\ln 3} \frac{\pi e^{2x} \sin \pi y^2}{y^2} dx dy dz$. | 07 |
| Q.3 (a) If an electrostatic field E acts on a liquid or a gaseous polar dielectric, the net dipole moment P per unit volume is $P(E) = \frac{e^E + e^{-E}}{e^E - e^{-E}} - \frac{1}{E}$. Show that $\lim_{E \rightarrow 0^+} P(E) = 0$. | 03 |
| (b) For what values of the constant k does the second derivative test guarantee that $f(x, y) = x^2 + kxy + y^2$ will have a saddle point at $(0,0)$? A local minimum at $(0,0)$? | 04 |
| (c) Find the series radius and interval of convergence for $\sum_{n=0}^\infty \frac{(3x-2)^n}{n}$. For what values of x does the series converge absolutely? | 07 |
| OR | |
| Q.3 (a) Determine whether the integral $\int_0^3 \frac{dx}{x-1}$ converges or diverges. | 03 |
| (b) Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 1, x = 4$ about the line $y = 1$. | 04 |

- (c) Check the convergence of the series $\sum_{n=1}^{\infty} \frac{(\ln n)^3}{n^3}$ and $\sum_{n=0}^{\infty} (-1)^n (\sqrt{n+1} - \sqrt{n})$. 07

- Q.4 (a) Show that the function $f(x, y) = \frac{2x^2y}{x^4+y^2}$ has no limit as (x, y) approaches to $(0,0)$. 03

- (b) Suppose f is a differentiable function of x and y and $g(u, v) = f(e^u + \sin v, e^u + \cos v)$. Use the following table to calculate $g_u(0,0), g_v(0,0), g_u(1,2)$ and $g_v(1,2)$. 04

	f	g	f_x	f_y
$(0,0)$	3	6	4	8
$(1,2)$	6	3	2	5

- (c) Find the Fourier series of 2π -periodic function $f(x) = x^2, 0 < x < 2\pi$ and hence deduce that $\frac{\pi^2}{6} = \sum_{n=0}^{\infty} \frac{1}{n^2}$. 07

OR

- Q.4 (a) Verify that the function $u = e^{-\alpha^2 k^2 t} \cdot \sin kx$ is a solution of the heat conduction equation $u_t = \alpha^2 u_{xx}$. 03

- (b) Find the half-range cosine series of the function $f(x) = \begin{cases} 2, & -2 < x < 0 \\ 0, & 0 < x < 2 \end{cases}$. 04

- (c) Find the points on the sphere $x^2 + y^2 + z^2 = 4$ that are closest to and farthest from the point $(3,1,-1)$. 07

- Q.5 (a) Find the directional derivative $D_u f(x, y)$ if $f(x, y) = x^3 - 3xy + 4y^2$ and u is the unit vector given by angle $\theta = \frac{\pi}{6}$. What is $D_u f(1,2)$? 03

- (b) Find the area of the region bounded by the curves $y = \sin x, y = \cos x$ and the lines $x = 0$ and $x = \frac{\pi}{4}$. 04

- (c) Prove that $A = \begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$ is diagonalizable and use it to find A^{13} . 07

OR

- Q.5 (a) Define the rank of a matrix and find the rank of the matrix $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 5 & -3 \\ 1 & -4 & 7 \end{bmatrix}$. 03

- (b) Use Gauss-Jordan algorithm to solve the system of linear equations $2x_1 + 2x_2 - x_3 + x_5 = 0$
 $-x_1 - x_2 + 2x_3 - 3x_4 + x_5 = 0$
 $x_1 + x_2 - 2x_3 - x_5 = 0$
 $x_3 + x_4 + x_5 = 0$ 04

- (c) State Cayley-Hamilton theorem and verify it for the matrix $A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER- I & II (NEW) EXAMINATION – WINTER 2019****Subject Code: 3110014****Date: 17/01/2020****Subject Name: Mathematics – I****Time: 10:30 AM TO 01:30 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.

		MARKS
Q.1	(a) Find the equations of the tangent plane and normal line to the surface $x^2 + y^2 + z^2 = 3$ at the point (1,1,1)	03
	(b) Evaluate $\lim_{x \rightarrow 0} \frac{xe^x - \log(1+x)}{x^2}$	04
	(c) Using Gauss Elimination method solve the following system $-x+3y+4z=30$ $3x+2y-z=9$ $2x-y+2z=10$	07
Q.2	(a) Test the convergence of the series $\frac{1}{3} + \left(\frac{2}{5}\right)^2 + \left(\frac{3}{7}\right)^3 + \dots + \left(\frac{n}{2n+1}\right)^n + \dots$	03
	(b) Discuss the Maxima and Minima of the function $3x^2 - y^2 + x^3$	04
	(c) Find the fourier series of $f(x) = \frac{(\pi-x)}{2}$ in the interval (0,2 π)	07
	OR	
	(c) Change the order of integration and evaluate $\int_0^1 \int_x^1 \sin y^2 dy dx$	07
Q.3	(a) Find the value of $\beta\left(\frac{7}{2}, \frac{5}{2}\right)$	03
	(b) Obtain the fourier cosine series of the function $f(x) = e^x$ in the range (0,l)	04
	(c) Find the maximum and minimum distance from the point (1,2,2) to the sphere $x^2 + y^2 + z^2 = 36$	07
	OR	
Q.3	(a) Test the convergence of the series $\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots$	03
	(b) Evaluate $\iint (x^2 - y^2) dx dy$ over the triangle with the vertices (0,1), (1,1), (1,2)	04
	(c) Find the volume of the solid generated by rotating the plane region bounded by $y = \frac{1}{x}$, $x=1$ and $x=3$ about the X axis.	07
Q.4	(a) Evaluate $\int_0^\pi \int_0^{\sin \theta} r dr d\theta$	03
	(b) Express $f(x) = 2x^3 + 3x^2 - 8x + 7$ in terms of (x-2)	04

- (c) Using Gauss-Jordan method find A^{-1} for $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ **07**

OR

- Q.4** (a) Using Cayley-Hamilton Theorem find A^{-1} for $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ **03**

- (b) Evaluate $\int_0^{\infty} \frac{dx}{x^2+1}$ **04**

- (c) Test the convergence of the series $\frac{x}{1 \cdot 2} + \frac{x^2}{3 \cdot 4} + \frac{x^3}{5 \cdot 6} + \frac{x^4}{7 \cdot 8} + \dots$ **07**

- Q.5** (a) Evaluate $\int_0^1 \int_1^2 xy \, dy \, dx$ **03**

- (b) Find the eigen values and eigenvectors of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$ **04**

- (c) If $u = f(x-y, y-z, z-x)$ then show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$ **07**

OR

- Q.5** (a) Find the directional derivatives of $f = xy^2 + yz^2$ at the point $(2, -1, 1)$ in the direction of $i+2j+2k$. **03**

- (b) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+1}$ **04**

- (c) Evaluate $\iiint xyz \, dx \, dy \, dz$ over the positive octant of the sphere $x^2 + y^2 + z^2 = 4$ **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II (NEW) EXAMINATION – WINTER 2020****Subject Code:3110014****Date:16/03/2021****Subject Name:Mathematics – I****Time:10:30 AM TO 12:30 PM****Total Marks:47****Instructions:**

1. Attempt any **THREE** questions from Q1 to Q6.
2. **Q7 is compulsory.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

	Marks
Q.1 (a) Expand $\sin x$ in powers of $(x - \pi/2)$.	03
(b) Evaluate $\lim_{x \rightarrow 0} \frac{\tan^2 x - x^2}{x^2 \tan^2 x}$.	04
(c) (i) Check the convergence of $\int_4^{\infty} \frac{3x+5}{x^4+7} dx$.	03
(ii) The region between the curve $y = \sqrt{x}$, $0 \leq x \leq 4$ and the line $x = 4$ is revolved about the x -axis to generate a solid. Find its volume.	04
Q.2 (a) If $u = \cos e^{-1} \left(\frac{x+y}{x^2+y^2} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$.	03
(b) Check the convergence of the series $\sum_{n=1}^{\infty} \frac{2^n + 5}{3^n}$.	04
(c) (i) Test the convergence of the series $\sum_{n=1}^{\infty} (\sqrt{n+1} - \sqrt{n})$.	03
(ii) Test the convergence of $\sum_{n=1}^{\infty} \frac{2^n}{n^3 + 1}$.	04
Q.3 (a) Solve the following equations by Gauss' elimination method: $x + y + z = 6, x + 2y + 3z = 14, 2x + 4y + 7z = 30$.	03
(b) If $u = f(x - y, y - z, z - x)$, prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.	04
(c) (i) Find the equation of the tangent plane and normal line to the surface $x^2 + 2y^2 + 3z^2 = 12$ at $(1, 2, -1)$.	03
(ii) For $f(x, y) = x^3 + y^3 - 3xy$, find the maximum and minimum values.	04
Q.4 (a) Find the rank of the matrix $\begin{bmatrix} 8 & 0 & 0 & 16 \\ 0 & 0 & 0 & 6 \\ 0 & 9 & 9 & 9 \end{bmatrix}$.	03
(b) If $u = f(x + at) + g(x - at)$, prove that $\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$.	04

- (c) (i) Show that the function $f(x, y) = \begin{cases} \frac{2x^2y}{x^3 + y^3}, (x, y) \neq (0, 0) \\ 0, (x, y) = (0, 0) \end{cases}$ is not continuous at the origin. 03

- (ii) Find the shortest distance from the point (1, 2, 2) to the sphere $x^2 + y^2 + z^2 = 16$. 04

- Q.5** (a) Use Gauss-Jordan method to find A^{-1} , if $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$. 03

- (b) Using Caley-Hamilton theorem find A^2 , if $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. Also find A^{-1} . 04

- (c) Find the Fourier cosine series for $f(x) = x^2, 0 < x < \pi$. Hence show that $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} = \frac{\pi^2}{12}$. 07

- Q.6** (a) Evaluate $\iint_R e^{2x+3y} dA$, where R is the triangle bounded by $x = 0, y = 0, x + y = 1$. 03

- (b) Find the eigen values and eigen vectors for the matrix $\begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$. 04

- (c) Evaluate $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dA$ by changing the order of integration. 07

- Q.7** Evaluate $\int_0^{\pi/2} \int_{a(1-\cos\theta)}^a r^2 dr d\theta$. 05

OR

- Q.7** Find the area enclosed within the curves $y = 2 - x$ and $y^2 = 2(2 - x)$. 05

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-1/2 EXAMINATION – WINTER 2021****Subject Code:3110014****Date:19/03/2022****Subject Name:Mathematics - 1****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	MARKS
Q.1 (a) If $u = \log(\tan x + \tan y + \tan z)$ then show that	03
$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$	
(b) Evaluate	04
$\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2\log(1+x)}{x \sin x}$	
(c) Find the extreme values of the function	07
$f(x, y) = x^3 + y^3 - 3x - 12y + 20$	
Q.2 (a) Use Ratio test to check the convergence of the series	03
$\sum_{n=1}^{\infty} \frac{2^n + 1}{3^n + 1}$	
(b) Find the Maclaurin's series of $\cos x$ and use it to find the series of $\sin^2 x$.	04
(c) Find the Fourier series of $f(x) = x^2$ in the interval $(0, 2\pi)$ and hence deduce that $\frac{\pi^2}{12} = \frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \dots$	07
OR	
(c) Find the Fourier series of $f(x) = 2x - x^2$ in the interval $(0, 3)$.	07
Q.3 (a) Find the directional derivative of $f(x, y, z) = xyz$ at the point $P(-1, 1, 3)$ in the direction of the vector $\vec{a} = \hat{i} - 2\hat{j} + 2\hat{k}$.	03
(b) Find the rank of the matrix $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$ by reducing to row echelon form.	04
(c) Find the eigenvalues and corresponding eigenvectors of the matrix	07
$A = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$	
OR	
Q.3 (a) If $u = f(x - y, y - z, z - x)$, then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$	03
(b) Find the inverse of the following matrix by Gauss-Jordan method:	04

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

- (c) Verify Cayley-Hamilton theorem for the following matrix and use it to find A^{-1} 07

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

- Q.4 (a)** If 1 is an eigenvalue of the matrix $\begin{bmatrix} 2 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ then find its corresponding eigen vector. 03

- (b) Expand $2x^3 + 7x^2 + x - 1$ in powers of $(x - 2)$ 04

- (c) Solve following system by using Gauss Jordan method 07

$$\begin{aligned} x + 2y + z - w &= -2 \\ 2x + 3y - z + 2w &= 7 \\ x + y + 3z - 2w &= -6 \\ x + y + z + w &= 2 \end{aligned}$$

OR

- Q.4 (a)** Use integral test to show that the following infinite series is convergent 03

$$\sum_{n=1}^{\infty} \frac{1}{n(1 + \log^2 n)}$$

- (b) For the odd periodic function defined below, find the Fourier series 04

$$f(x) = \begin{cases} -1, & -1 < x < 0 \\ 1, & 0 < x < 1 \end{cases}$$

- (c) Determine the radius and interval of convergence of the following infinite series 07

$$x - \frac{x^2}{4} + \frac{x^3}{9} - \frac{x^4}{16} + \dots$$

- Q.5 (a)** Show the following limit does not exist using different path approach 03

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y^2}{x^4 + y^4}$$

- (b) Evaluate the following integral along the region R 04

$$\iint_R (x + y) dy dx$$

where R is the region bounded by $x = 0, x = 2, y = x, y = x + 2$. Also, sketch the region.

- (c) Change the order of integration and hence evaluate the same. Do sketch the region. 07

$$\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx$$

OR

- Q.5 (a)** The following integral is an improper integral of which type? Evaluate 03

$$\int_0^{\infty} \frac{dx}{x^2 + 1}$$

- (b) If $x = r\sin\theta\cos\varphi, y = r\sin\theta\sin\varphi, z = r\cos\theta$, then find the jacobian 04

$$\frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)}$$

- (c) Find the volume of the solid generated by rotating the region bounded by $y = x^2 - 2x$ and $y = x$ about the line $y = 4$. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW)EXAMINATION – SUMMER 2022****Subject Code:3110014****Date:02-08-2022****Subject Name:Mathematics - 1****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Is $\sum_{n=1}^{\infty} \frac{1}{n^p}$ convergent for $p > 1$? Justify your answer.	03
(b) (1) Find $\lim_{x \rightarrow a} \frac{\sin x - \sin a}{(x-a)^2}$	02
(2) Is $\int_0^{\infty} \frac{\sin^2 x}{x^2} dx$ convergent? Justify your answer.	02
(c) (1) Find the length of curve $f(x) = \frac{x^3}{12} + \frac{1}{x}, 1 \leq x \leq 4.$	04
(2) Prove that $\Gamma(n) = (n-1) \Gamma(n-1).$	03
Q.2 (a) Investigate the convergence of $\sum_{n=1}^{\infty} \frac{n^2}{7^n}.$	03
(b) Investigate the convergence of $\sum_{n=1}^{\infty} \frac{2^n (n!)^2}{(2n)!}$	04
(c) Find Fourier series of $f(x) = x^2, -\pi < x < \pi.$	07
OR	
(c) Find Fourier series of $f(x) = \begin{cases} x, & -1 < x < 0 \\ 2, & 0 < x < 1 \end{cases}.$	07
Q.3 (a) Find the derivative of $f(x, y) = x^2 + xy + y^2$ in the direction $\hat{i} + \hat{j}$ at $P(1,1).$	03
(b) Find the tangent plane of $z = e^x \cos y$ at $P(0,0,0).$	04
(c) Find local extreme values of $f(x, y) = xy - x^2 - y^2 - x.$	07
OR	
Q.3 (a) Explain second derivative test for local extreme values.	03
(b) Let $f = \ln r$, where $\vec{r} = x \hat{i} + y \hat{j} + z \hat{k}$ and $r = \vec{r} $. Find $\text{grad } f$.	04
(c) Determine the minimum value of $x^2 y z^2$ subject to the condition $x + y + 2z = 5$ using method of Lagrange multipliers.	07
Q.4 (a) Evaluate $\int_{y=0}^1 \int_{x=0}^2 \frac{1}{\sqrt{4-x^2} \sqrt{1-y^2}} dx dy.$	03

- (b) Evaluate the integral $\int_0^2 \int_{x/2}^1 \frac{1}{3} e^{y^2} dy dx$ 04

by change of order.

- (c) (1) Find the area of the region covered by $x = 1$, $x = 4$, $y = 0$ and $y = \sqrt{x}$. 04

- (2) Evaluate $\int_{x=0}^1 \int_{y=0}^{x^{1/4}} \int_{z=0}^{y^2} \sqrt{z} dz dy dx$ 03

OR

- Q.4 (a)** Evaluate $\iint_R xy dA$ where R is the region 03

bounded by x axis, the ordinate $x = 2a$ and the curve $x^2 = 4ay$.

- (b) Evaluate the integral $\int_{y=0}^1 \int_{x=0}^{\cos^{-1} y} e^{\sin x} dx dy$ by change of order. 04

- (c) (1) Change in to polar coordinates then solve $\int_0^2 \int_0^{\sqrt{4-x^2}} e^{-(x^2+y^2)} dy dx$. 04

- (2) Let $x + y = u$ and $y = uv$ are given transformations. Find Jacobian for change of variables. 03

- Q.5 (a)** Find characteristic equation of $A = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 2 & 1 \\ 2 & 0 & 1 \end{bmatrix}$. 03

- (b) Find Maclaurin's series for $f(x) = e^{2x} \sinh x$ and show at least up to x^4 term. 04

- (c) Solve $x + y + w = 1$, $2x + z + w = 3$, $2y + z + 2w = 2$. 07

OR

- Q.5 (a)** Show that give matrix $A = \begin{bmatrix} 0 & 1 \\ -2 & -1 \end{bmatrix}$ satisfies its Characteristic equation. 03

- (b) Show that $\sum_{n=2}^{\infty} \frac{(-1)^n}{\ln n}$ converges. 04

- (c) Show that $A = \begin{bmatrix} 0 & 1 & 0 \\ -2 & -1 & 2 \\ -4 & -8 & 7 \end{bmatrix}$ is diagonalizable. Find the matrix of eigen vectors and diagonal matrix. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-I & II(NEW) EXAMINATION – WINTER 2022****Subject Code:3110014****Date:02-03-2023****Subject Name:Mathematics - 1****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- Q.1** (a) Find $\lim_{x \rightarrow 0} \frac{\tan^2 x - x^2}{x^2 \tan^2 x}$. **03**
- (b) If $z = \tan^{-1} \frac{x}{y}$, $x = u \cos v$ and $y = u \sin v$ then find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$ using the chain rule. **04**
- (c) Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3xy$. **07**
- Q.2** (a) Find the directional derivative of the function $f(x, y) = x^2 \sin 2y$ at the point $(1, \frac{\pi}{2})$ in the direction of $v = 3\hat{i} - 4\hat{j}$ **03**
- (b) Show that the series $\sum_{n=0}^{\infty} \left(\frac{e}{\pi}\right)^n$ is convergent and find its sum. **04**
- (c) Find the Fourier series of the 2π periodic function $f(x) = x + |x|$, $-\pi < x \leq \pi$. **07**
- OR**
- (c) Find the half range sine series of the function $f(x) = x - x^2$ in the interval $(0, 1)$ and hence determine the sum $\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^3}$. **07**
- Q.3** (a) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{2n+1}{n^2+n+1}$. **03**
- (b) Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & -1 & 0 \\ -1 & 3 & 0 & -4 \\ 2 & 1 & 3 & -2 \\ 1 & 1 & 1 & -1 \end{bmatrix}$. **04**
- (c) Solve the following system of linear equations using the Gauss-Jordan Method. **07**
 $x_1 + x_2 + 2x_3 = 8, -x_1 - 2x_2 + 3x_3 = 1, 3x_1 - 7x_2 + 4x_3 = 10.$
- OR**
- Q.3** (a) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{(n+1) \sin\left(\frac{(2n-1)\pi}{2}\right)}{n \log(n+1)}$ **03**
- (b) Find the approximate value of $\sqrt{25.15}$ using Taylor's theorem. **04**
- (c) Find the eigenvalues and the eigenvectors of the matrix $A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$. **07**
- Q.4** (a) Let $A = \begin{bmatrix} -3 & -1 & 1 \\ 0 & 2 & 3 \\ 1 & -2 & 1 \end{bmatrix}$. Using Cayley-Hamilton theorem, find the matrix A^3 . **03**
- (b) Find the equations of the tangent plane and normal line to the surface $x^2 y z + 3y^2 = 2xz^2 - 8z$ at $P(1, 2, -1)$. **04**

- (c) Find the length of the arc of the curve $y = \log \left(\frac{e^x - 1}{e^x + 1} \right)$ from $x = 1$ to $x = 2$. 07

OR

- Q.4** (a) Show that $f(x, y) = \begin{cases} \frac{x^3 y}{x^4 + y^4}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$ is not continuous at $(0, 0)$. 03

- (b) Find the inverse of the matrix $A = \begin{bmatrix} 1 & 3 & 3 \\ 1 & 4 & 3 \\ 1 & 3 & 4 \end{bmatrix}$ using Gauss-Jordan method. 04

- (c) Sketch the region R bounded by the lines $x = 0, x = 2, x = y$ and $y = x + 2$. Find the area of this region R by double integrals. 07

- Q.5** (a) Evaluate $\int_0^1 \int_x^1 \int_0^{y-x} dz dy dx$. 03

- (b) Evaluate $\iint_R y(x - y^2) dy dx$ 04

where R is the region bounded by the curves $y = \sqrt{x}$ and $y = x^3$.

- (c) Determine all the positive values of x for which the series 07

$$\frac{1}{1 \cdot 2 \cdot 3} + \frac{x}{4 \cdot 5 \cdot 6} + \frac{x^2}{7 \cdot 8 \cdot 9} + \dots$$

converges.

OR

- Q.5** (a) Evaluate $\iint_R xy \sqrt{x^2 + y^2} dx dy$ by changing into polar coordinates where 03

$R = \{(x, y): 1 \leq x^2 + y^2 \leq 4, x \geq 0, y \geq 0\}$.

- (b) Discuss the convergence of the improper integral $\int_1^\infty \frac{\log x}{x^2} dx$. 04

- (c) Evaluate $\int_0^1 \int_{1-\sqrt{1-y^2}}^{1+\sqrt{1-y^2}} dx dy$ by changing the order of integration. Give the sketch of region of integration. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II(NEW) EXAMINATION – SUMMER 2023****Subject Code:3110014****Date:31-07-2023****Subject Name:Mathematics - 1****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Evaluate: $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$.	03
	(b) Show that series $\sum_{n=0}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right)$ is convergent. Hence find sum of the series.	04
	(c) Define eigenvalue and eigen vector of a matrix. Find all eigenvalues and eigenvectors of matrix $A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$.	07
Q.2	(a) Evaluate: $\int_0^1 \int_x^{2x} \int_0^y 2xyz \, dz \, dy \, dx$.	03
	(b) Find equations of tangent plane and normal line to the surface $\frac{x^2}{2} - \frac{y^2}{3} - z = 0$ at $(2, 3, -1)$.	04
	(c) Find Fourier series of 2π -periodic function $f(x) = x + x^2; -\pi < x < \pi$.	07
	OR	
	(c) Find the Fourier series for the periodic extension of $f(x) = \begin{cases} \sin x, & 0 \leq x \leq \pi \\ 0, & \pi \leq x \leq 2\pi \end{cases}$.	07
Q.3	(a) Define Improper integrals of type-I. Evaluate $\int_{-1}^{\infty} \frac{dt}{t^2 + 5t + 6}$.	03
	(b) Find the volume of the solid generated by revolving the region bounded by $y = x^2, y = 0$ and $x = 2$ about the x -axis.	04
	(c) (i) Using ratio test discuss convergence of series $\sum_{n=1}^{\infty} \frac{(n+3)!}{3!n!3^n}$.	03
	(ii) Use Taylor series to represent function $f(x) = 2x^3 + x^2 + 3x - 8$ in powers of $(x - 1)$.	04

OR

- Q.3 (a)** Define Beta function. Evaluate $\int_0^1 x^4(1-\sqrt{x})^5 dx$. **03**
- (b)** Find the arc length of $f(x) = \ln(\cos x)$, $0 \leq x \leq \pi/4$. **04**
- (c) (i)** Using Sandwich theorem find limit of sequence $(an) = \left(\frac{\cos n}{n}\right)$. **03**
- (ii)** Define Radius of Convergence of power series. Find it for power series $\sum_{n=1}^{\infty} \frac{(x-1)^n}{n^3 3^n}$. **04**
- Q.4 (a)** Find the sum of series $\sum_{n=0}^{\infty} \frac{2^n - 1}{3^n}$. **03**
- (b)** If $u = \log(x^2 + y^2)$, show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$. **04**
- (c)** Find all local maxima, local minima and saddle points of the function $f(x, y) = x^3 + 3xy^2 - 15x + y^3 - 15y$. **07**
- OR**
- Q.4 (a)** Discuss Convergence of series $\sum_{n=1}^{\infty} \ln\left(\frac{n}{2n+1}\right)$. **03**
- (b)** Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - y^2}{x^4 + y^2}$ does not exist. **04**
- (c)** A rectangular box without a lid is to be made from 12 m² of cardboard. Find the maximum volume of such a box. **07**
- Q.5 (a)** Find Jacobian $\frac{\partial(x, y)}{\partial(r, \theta)}$ for $x = r \cos \theta$, $y = r \sin \theta$. **03**
- (b)** Solve the given linear system using Gauss-Elimination method $x_1 + x_2 - 2x_3 = 1$, $2x_1 - 3x_2 + x_3 = -8$, $3x_1 + x_2 + 4x_3 = 7$. **04**
- (c)** Sketch the region of integration, change the order of integration and evaluate $\int_0^4 \int_{\sqrt{x}}^2 \frac{1}{y^3 + 1} dy dx$. **07**
- OR**
- Q.5 (a)** Evaluate: $\int_0^2 \int_0^{\pi/2} x \sin y dy dx$. **03**
- (b)** State Caley-Hamilton theorem. Verify it for matrix $A = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$. **04**
- (c)** Use the transformation $x = u^2 - v^2$, $y = 2uv$ to evaluate the integral $\int_0^1 \int_0^{2\sqrt{1-x}} \sqrt{x^2 + y^2} dy dx$. **07**

Seat No.: _____

Enrolment No. _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I (NEW) EXAMINATION – WINTER 2023

Subject Code:3110014

Date:11-01-2024

Subject Name:Mathematics - 1

Time:02:30 PM TO 05:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Evaluate $\lim_{x \rightarrow 0^+} x \ln x$	03
	(b) Define beta and gamma functions. What is the relationship between beta and gamma functions?	04
	(c) Solve the following system of linear equations using Gauss-Jordan elimination:	07
	$\begin{aligned} x_3 + x_4 + x_5 &= 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \\ x_1 + x_2 - 2x_3 - x_5 &= 0 \\ 2x_1 + 2x_2 - x_3 + x_5 &= 0 \end{aligned}$	
Q.2	(a) Define rank of the matrix. Find rank(A) if $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$	03
	(b) Test the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{1}{1+n}\right)^n$	04
	(c) Find eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$	07
	OR	
	(c) Find the Fourier series of the function $f(x) = x + \pi$ if $-\pi < x < \pi$ and $f(x + 2\pi) = f(x)$	07
Q.3	(a) If $w = x^2 + y^2, x = r - s, y = r + s$, using chain rule, prove that $\frac{\partial w}{\partial s} = 4s$.	03
	(b) Find the directional derivative of $f(x, y, z) = 2x^2 + 3y^2 + z^2$ at point (2,1,3) in the direction of the vector $\mathbf{i} - 2\mathbf{k}$.	04
	(c) Find local extreme values of the function $f(x, y) = 4x^2 + 9y^2 + 8x - 36y + 24$.	07
	OR	
Q.3	(a) Determine whether $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y}{x^3+y^3}$ exists and find it if exists.	03
	(b) Find the equation of the tangent plane to $z = 3x^2 - xy$ at the point (1,2,1).	04

- (c) Find the maximum and minimum values of the function $f(x, y) = 3x + 4y$ on the circle $x^2 + y^2 = 1$. 07
- Q.4** (a) Calculate $\iint_R f(x, y) dA$ for $f(x, y) = 100 - 6x^2y$ and $R: 0 \leq x \leq 2, -1 \leq y \leq 1$ 03
- (b) Sketch the region of integration for the integral $\int_0^2 \int_{x^2}^{2x} (4x + 2) dy dx$ and write an equivalent integral with the order of integration reversed. 04
- (c) Calculate $\iint_R \frac{\sin x}{x} dA$ where R is the triangle in the xy -plane bounded by the x -axis, the line $y = x$ and the line $x = 1$. 07
- OR**
- Q.4** (a) Evaluate $\int_1^{\ln 8} \int_0^{\ln y} e^{x+y} dx dy$. 03
- (b) Find the area of the region R bounded by $y = x$ and $y = x^2$ in the first quadrant. 04
- (c) Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2) dy dx$ by changing into polar coordinates. 07
- Q.5** (a) Find the Maclaurin series for $\cos x$. 03
- (b) Determine the convergence or divergence of the series $\sum_{n=1}^{\infty} n e^{-n^2}$. 04
- (c) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$. 07
- OR**
- Q.5** (a) Define monotonic sequence. Is the sequence $\left\{\frac{1}{n^2}\right\}$ monotonic? 03
- (b) Investigate the convergence of the series $\sum_{n=0}^{\infty} \frac{2^n + 5}{3^n}$. 04
- (c) Find the interval of convergence of the series $x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ 07

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER-I & II (NEW) EXAMINATION – SUMMER 2024

Subject Code:3110014

Date:09-07-2024

Subject Name:Mathematics - 1

Time:02:30 PM TO 05:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Q.1 (a) Using L' Hospital's rule, evaluate $\lim_{x \rightarrow 1} \frac{x - x^x}{1 + \log x - x}$. 03

(b) Define Beta function and evaluate $\int_0^1 x^3 (1 - \sqrt{x})^5 dx$. 04

(c) Find the Fourier series of $f(x) = \begin{cases} \pi + x & -\pi < x < 0 \\ \pi - x & 0 < x < \pi \end{cases}$ 07

Q.2 (a) Show that the sequence $\{u_n\}$, where $u_n = \frac{\sin n}{n}$ converges to zero. 03

(b) Express $f(x) = 2x^3 + 3x^2 - 8x + 7$ in terms of $(x - 2)$. 04

(c) Find the area of the surface of revolution of a quadrant of a circular arc as obtained by revolving it about a tangent at one of its ends. 07

OR

(c) Find the length of the loop of the curve $9ay^2 = (x - 2a)(x - 5a)^2$. 07

Q.3 (a) Evaluate $\int_0^\infty \frac{dv}{(1 + v^2)(1 + \tan^{-1} v)}$. 03

(b) Test the convergence of the series $\sum_{n=1}^\infty \frac{2n+1}{n^2(n+1)^2}$. 04

(c) If $\theta = t^n e^{-\frac{r^2}{4t}}$ then find n so that $\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \theta}{\partial r} \right) = \frac{\partial \theta}{\partial t}$. 07

OR

Q.3 (a) Check the convergence of $\int_0^5 \frac{1}{x^2} dx$. 03

(b) Evaluate $\int_0^\infty 3^{-4x^2} dx$. 04

Find the Fourier series of $f(x) = \frac{1}{2}(\pi - x)$ in the interval $(0, 2\pi)$. Hence, deduce 07

(c) that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$

- Q.4 (a)** Prove that $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$ **03**
- (b)** Find the Fourier sine series of $f(x) = e^x$ in $0 < x < \pi$. **04**
- (c)** Find the extreme values of the function $f(x, y) = x^3 + y^3 - 3x - 12y + 20$. **07**
- OR
- Q.4 (a)** Find the directional derivative of $f(x, y, z) = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction of $2\hat{i} - \hat{j} - 2\hat{k}$. **03**
- Solve the following system by Gauss – Jordan method: **04**
- $-2y + 3z = 1$
- (b)** $3x + 6y - 3z = -2$
- $6x + 6y + 3z = 5$
- Change the order of integration and evaluate **07**
- (c)**
$$\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{\cos^{-1} x}{\sqrt{1-x^2} \sqrt{1-x^2-y^2}} dx dy.$$
- Q.5 (a)** Evaluate $\int_0^3 \int_0^1 (x^2 + 3y^2) dy dx$. **03**
- (b)** Apply Cayley – Hamilton theorem to $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ and deduce that $A^8 = 625I$. **04**
- (c)** Evaluate $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x}{x^2 + y^2} dy dx$ by changing to polar coordinates. **07**
- OR
- Q.5 (a)** Evaluate $\int_0^2 \int_1^2 \int_0^{yz} xyz dx dy dz$. **03**
- (b)** Using Gauss Jordan method, find inverse of $A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$. **04**
- (c)** Find the eigen values and eigen vectors of the matrix $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$. **07**

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

Subject Code:3110014

Date:03-01-2025

Subject Name:Mathematics - 1

Time:10:30 AM TO 01:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Q.1 (a) Find the equation of tangent plane and normal line to the surface $xyz = 6$ at $(1, 2, 3)$. **03**

(b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$. **04**

(c) Solve the following system by Gauss Elimination method: **07**
$$\begin{aligned} x + y + 2z &= 9 \\ 2x + 4y - 3z &= 1 \\ 3x + 6y - 5z &= 0 \end{aligned}$$

Q.2 (a) Find the directional derivative of $f(x, y, z) = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction $\hat{i} + 2\hat{j} + 2\hat{k}$. **03**

(b) Expand $x^4 - 3x^3 + 2x^2 - x + 1$ in powers of $(x - 3)$. **04**

(c) Find the Fourier series of $f(x) = \begin{cases} x^2 & 0 < x < \pi \\ 0 & \pi < x < 2\pi \end{cases}$. **07**

OR

(c) Find the Fourier series of $f(x) = x + |x|$ in the interval $-\pi < x < \pi$. **07**

Q.3 (a) Show that the sequence $\{u_n\}$, where $u_n = \frac{\sin n}{n}$ converges to zero. **03**

(b) Evaluate $\int_0^3 \frac{dx}{(x-1)^{\frac{2}{3}}}$. **04**

(c) A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such box. **07**

OR

Q.3 (a) Evaluate $\int_0^1 (\log x)^5 dx$. **03**

(b) Find the cosine series for $f(x) = \pi - x$ in the interval $0 < x < \pi$. **04**

(c) Find the extreme values of the function $x^3 + xy^2 + 21x - 12x^2 - 2y^2$. **07**

Q.4 (a) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 1}$. **03**

- (b) Evaluate the integral $\int_0^{\frac{\pi}{2}} \int_0^{1-\sin\theta} r^2 \cos\theta \, dr \, d\theta$. 04
- (c) Find the length of the parabola $x^2 = 4y$ which lies inside the circle $x^2 + y^2 = 6y$ 07
- OR**
- Q.4** (a) Evaluate $\int_0^2 \int_1^z \int_0^{yz} xyz \, dx \, dy \, dz$. 03
- (b) Test the convergence of the series $\sum \frac{(n+1)^n x^n}{n^{n+1}}$. 04
- (c) Change the order of integration and evaluate $\int_0^\infty \int_x^\infty \frac{e^{-y}}{y} \, dy \, dx$. 07
- Q.5** (a) Test the convergence of the series $\sum_{n=1}^\infty n^2 e^{-n^3}$. 03
- (b) Find a and b such that $A = \begin{bmatrix} a & 4 \\ 1 & b \end{bmatrix}$ has 3 and -2 as eigenvalues. 04
- (c) If $u = f\left(\frac{y-x}{xy}, \frac{z-x}{xz}\right)$, show that $x^2 \frac{\partial u}{\partial x} + y^2 \frac{\partial u}{\partial y} + z^2 \frac{\partial u}{\partial z} = 0$. 07
- OR**
- Q.5** (a) Evaluate $\int_0^\infty \frac{dv}{(1+v^2)(1+\tan^{-1} v)}$. 03
- (b) Using Gauss Jordan method find inverse of $\begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$. 04
- (c) Show that the matrix $\begin{bmatrix} 1 & -2 & 0 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ is not diagonalizable. 07

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:3110014****Date:24-06-2025****Subject Name:Mathematics - 1****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

- | | MARKS |
|---|-----------|
| Q.1 (a) Evaluate: $\lim_{x \rightarrow 0} [x^{-2} - \operatorname{cosec}^2 x]$. | 03 |
| (b) Check the consistency of the system of linear equations. Solve it if consistent.
$\begin{aligned} 3x + y - 3z &= 13, \\ 2x - 3y + 7z &= 5, \\ 2x + 19y - 47z &= 32. \end{aligned}$ | 04 |
| (c) Find the Fourier Series of the function $f(x) = x^2$ in the interval $(-\pi, \pi)$. | 07 |
| Q.2 (a) Define the improper integrals of the first kind and the second kind. State the relation between Beta and Gamma function. | 03 |
| (b) Find the area of the surface of revolution generated by revolving the curve $x = y^3$ from $y = 0$ to $y = 2$. | 04 |
| (c) Find eigen values and corresponding eigen vectors of the matrix $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$. | 07 |
| OR | |
| (c) Find the inverse of the matrix using Gauss-Jordan elimination method.
$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 0 \\ 0 & 1 & 2 \end{pmatrix}$ | 07 |
| Q.3 (a) Determine whether $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ exists and find it if exists. | 03 |
| (b) Find the equation of tangent plane and normal line to the surface $2xz^2 - 3xy - 4x = 7$ at the point $(1, -1, 2)$. | 04 |
| (c) Find the extreme values of the function $x^3 + y^3 - 63(x + y) + 12xy$. | 07 |
| OR | |
| Q.3 (a) State chain rule for $\frac{\partial u}{\partial x}$ for $u = f(v, w)$, where $v = g(x, y)$, $w = h(x, y)$.
If $u = f(x - 2y, 2y - 3z, 3z - x)$ then show that $\frac{\partial u}{\partial x} + \frac{1}{2} \frac{\partial u}{\partial y} + \frac{1}{3} \frac{\partial u}{\partial z} = 0$. | 03 |
| (b) Find the directional derivative of $f(x, y, z) = xy^2 + yz^2$ at the point $(2, -1, 1)$ in the direction of the vector $\hat{i} + 2\hat{j} + 2\hat{k}$. | 04 |

- (c) The temperature $T(x, y, z)$ at any point in space is $T = 400xyz^2$. Find the highest temperature on surface of the sphere $x^2 + y^2 + z^2 = 1$. 07

Q.4 (a) Evaluate $\int_{-1}^1 \int_0^2 (1 - 6x^2y) dx dy$. 03

- (b) Find the value of $\iint_R (2x - y^2) dA$ over the triangular region R enclosed between the line $y = -x + 1$, $y = x + 1$ and $y = 3$. 04

(c) Change the order of integral and evaluate $\int_0^{4a} \int_{x^2/4a}^{2\sqrt{ax}} dy dx$, ($a > 0$). 07

OR

Q.4 (a) Calculate $\int_0^2 \int_1^z \int_0^{yz} xyz dx dy dz$. 03

(b) Evaluate $\int_0^1 \int_x^1 \sin y^2 dy dx$. 04

(c) Compute $\int_0^2 \int_0^{\sqrt{2x-x^2}} \frac{x}{x^2+y^2} dy dx$ by transforming into polar coordinates. 07

Q.5 (a) Define Monotonic sequence. Test the convergence of the sequence $\{2 - (-1)^n\}$. 03

(b) Express the function $f(x) = \log(1 + x)$ in power series using the formula of Maclaurin's series. 04

(c) (i) Test the convergence of $\sum_{n=1}^{\infty} \frac{n!}{(2n+1)!}$. 07

(ii) State Cauchy's root test and discuss the convergence of $\sum_{n=1}^{\infty} \frac{1}{(\log n)^n}$.

OR

Q.5 (a) State sandwich theorem for sequence. Show that the sequence $u_n = \frac{\sin n}{n}$ converges to zero. 03

(b) Using Taylor's theorem find the approximate value of $\sqrt{10}$ up to three decimal places. 04

(c) (i) Examine the convergence of the series $\frac{1}{1 \cdot 2} - \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} - \frac{1}{5 \cdot 6} + \frac{1}{7 \cdot 8} - \dots \dots \dots$ 07

(ii) State Cauchy's Integral test for convergence of series and test the convergence of the series $\sum_{n=1}^{\infty} n^2 e^{-n^3}$.

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2025****Subject Code:3110014****Date:17-01-2026****Subject Name: Mathematics - 1****Time:02:30 PM TO 05:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Evaluate $\lim_{x \rightarrow 0} (\cos x)^{\cot x}$	3
(b) Discuss the convergence of the series $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \frac{1}{4 \cdot 5} + \dots$. If convergent find its sum.	4
(c) 1. Investigate the convergence of the integral $\int_0^1 \frac{1}{1-x} dx$.	3
2. Find the length of the curve $(x) = \frac{x^3}{12} + \frac{1}{x}$; $1 \leq x \leq 4$.	4
Q.2 (a) Discuss the convergence of the series $\sum_{n=1}^{\infty} \frac{2n+1}{(n+1)^2}$	3
(b) Discuss the convergence of the following: 1. $\sum_{n=1}^{\infty} \frac{4^n + 5^n}{6^n}$ 2. $\sum_{n=1}^{\infty} \frac{2^n}{n^3 + 1}$	4
(c) Obtain the Fourier series of the function $f(x) = \begin{cases} 0 & \text{if } -\pi < x \leq 0 \\ x & \text{if } 0 \leq x < \pi \end{cases}$	7
OR	
(c) Obtain the Fourier series of the function $f(x) = e^{ x }$; $-2 \leq x \leq 2$	7
Q.3 (a) If $u = f(x-y, y-z, z-x)$, Show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.	3
(b) Find the tangent plane and normal line of the surface $x^3 + y^3 + z^3 = 3$ at the point $(1,1,1)$.	4
(c) Find the extreme values that the function $f(x, y) = xy$ takes on the ellipse $\frac{x^2}{8} + \frac{y^2}{2} = 1$.	7
OR	
Q.3 (a) $u = 2x^3y + y^3z^2$, where $x = rse^t$, $y = rs^2e^{-t}$ and $z = r^2s \sin t$, find $\frac{\partial u}{\partial s}$ at $r = 1, s = 1, t = 0$.	3
(b) If $u = \tan^{-1} \left(\frac{x^2 + y^2}{x-y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \sin 2u$.	4
(c) Examine the function $f(x, y) = 2x^4 + y^2 - x^2 - 2y$ for maxima and minima.	7
Q.4 (a) Evaluate $\int_0^1 \int_0^x e^{\frac{y}{x}} dy dx$.	3
(b) Evaluate $\iint_R f(x, y) dA$, where $f(x, y) = 6x^2 + 2y$ and R is the region bounded by $y = x^2$ and $y = 4$.	4

- (c) By using changing the order of integration, evaluate $\int_0^1 \int_{-\sqrt{1-y^2}}^{1-y} y \, dx \, dy$. 7

OR

- Q.4 (a) Evaluate $\int_0^{\pi/2} \int_0^{a \cos \theta} r \sin \theta \, dr \, d\theta$. 3
 (b) Evaluate $\int_0^1 \int_0^{1-y} \int_0^{1-y-z} z \, dx \, dz \, dy$. 4
 (c) Evaluate $\iint_R x \, dx \, dy$, where R is the region bounded by triangle with vertices $(0,0)$, $(0,1)$ and $(1,1)$, using the transformations $x = u$, $y = uv$. 7

- Q.5 (a) Verify Cayley-Hamilton theorem for the matrix $\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. 3
 (b) Expand $\ln(1 + \sin x)$ in powers of x up to x^3 . 4
 (c) Solve the following system using Gauss-Jordan method. 7

$$\begin{aligned} x - 2y - z + 3w &= 1 \\ 2x - 4y + z &= 5 \\ x - 2y + 2z - 3w &= 4 \end{aligned}$$

OR

- Q.5 (a) Using row echelon form find the rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & -1 \\ 3 & 1 & 1 \end{bmatrix}$. 3
 (b) Discuss the convergence of the series $\frac{1}{1+3} + \frac{2}{1+3^2} + \frac{3}{1+3^3} + \frac{4}{1+3^4} + \dots$. 4
 (c) Find the modal matrix P and diagonal matrix D for the matrix $A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$. 7
